Xuan Luo

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Education

University of Washington | ***PhD Student in Computer Science & Engineering*** Sept. 2015 – 2022
Advisors: Steven M. Seitz, Jason Lawrence, and Ricardo Martin-Brualla. Seattle, WA

Shanghai Jiao Tong University | ***BS in Computer Science & Technology*** Sept. 2011 – June 2015
ACM Honors Class (one of the top gifted CS programs in China). Shanghai, China

Areas of Expertise

Long Context Modeling, Efficient Transformer, 3D Vision, Image Synthesis, Computational Photography

Professional Experience

Google | ***Research Scientist*** 2022 – Present
Project Beam —– the future of communications. Seattle, WA

- Drove the core development of a real-time 3D communication system, successfully demonstrating the importance and feasibility of temporal novel view synthesis systems for next-generation products, setting a new technical direction for the team.
- Integrating 3D geometry, enabled transformers to also achieve real-time, 4K novel view synthesis.
- Designed long-context modeling mechanisms, balancing long-term memorization with real-time constraints.
- Developed sparse attention algorithms to enable high-resolution novel view synthesis at real-time speeds.
- Co-authored *Quark*, a real-time, high-resolution, and generalizable neural view synthesis framework (**Best Paper, SIGGRAPH Asia 2024**).
- Developed flow-based and diffusion-based generative models for photorealistic 3D avatars.

University of Washington | ***Graduate Student Researcher*** 2015 – 2022
Advisors: Steven M. Seitz, Jason Lawrence, and Ricardo Martin-Brualla Seattle, WA

- Restored what famous historical figures would look like if rephotographed with modern cameras.
- Collected a large-scale rectified historical stereo dataset and visualized historical scenes in 3D.
- Designed an inexpensive glass-free DIY 3D display with a tablet and a plastic sheet folded into a cone.

Facebook | ***Research Intern*** June 2019 – March 2020
Mentor: Johannes Kopf. Collaborators: Jia-bin Huang, Kevin Matzen, and Richard Szeliski Seattle, WA

- Estimated geometrically consistent depth from monocular videos, enabling video effects to a whole new level.

Disney Research Zurich | ***Research Intern*** Summer 2017
Collaborators: Thabo Beeler, Derek Bradley, Matthias Niessner, and Paulo Gotardo. Zurich, Switzerland

- Worked on facial motion capture.

Google Daydream | ***Software Engineering Intern*** Summer 2016
Mentor: Jason Lawrence Seattle, WA

National University of Singapore | ***Visiting Scholar*** Aug. 2014 – Feb. 2015
Advisor: Shuicheng Yan Singapore

Publications

Online Neural Space Time Memory for Dynamic Novel View Synthesis 2026
Baback Elmieh, Lynn Tsai, Zeman Li, Srinivas Kaza, Tiancheng Sun, Gabor Csapo,
Ali Behrouz, Yuan Deng, Stephen Lombardi, Steven M. Seitz, **Xuan Luo**

Scaling Transformer-Based Novel View Synthesis with Models Token Disentanglement and Synthetic Data ICCV 2025
Nithin Gopalakrishnan Nair, Srinivas Kaza, **Xuan Luo**, Vishal M. Patel, Stephen Lombardi,
Jungyeon Park

Learning Efficient Fuse-and-Refine for Feed-Forward 3D Gaussian Splatting NeurIPS 2025
Yiming Wang, Lucy Chai, **Xuan Luo**, Michael Niemeyer, Manuel Lagunas,
Stephen Lombardi, Siyu Tang, Tiancheng Sun

LVT: Large-Scale Scene Reconstruction via Local View Transformers SIGGRAPH Asia 2025
Tooba Imtiaz*, Lucy Chai*, Kathryn Heal, **Xuan Luo**, Jungyeon Park, Jennifer Dy, John Flynn

Quark: Real-time, High-resolution, and General Neural View Synthesis SIGGRAPH Asia 2024
Best Paper Award
John Flynn*, Michael Broxton*, Lukas Murmann*, Lucy Chai, Matthew DuVall,
Clément Godard, Kathryn Heal, Srinivas Kaza, Stephen Lombardi, **Xuan Luo**,
Supreeth Achar, Kira Prabhu, Tiancheng Sun, Lynn Tsai, Ryan Overbeck

StyleSDF: High-Resolution 3D-Consistent Image and Geometry Generation CVPR 2022
Roy Or-El, **Xuan Luo**, Mengyi Shan, Eli Shechtman, Jeong Joon Park, Ira Kemelmacher-Shlizerman

Time-Travel Rephotography | SIGGRAPH Asia TOG 2021
Xuan Luo, Cecilia Zhang, Paul Yoo, Ricardo Ricardo Martin-Brualla, Jason Lawrence, and Steven M. Seitz

Consistent Video Depth Estimation | SIGGRAPH TOG 2020
Xuan Luo, Jia-Bin Huang, Richard Szeliski, Kevin Matzen, and Johannes Kopf

KeystoneDepth: History in 3D | International Virtual Conference on 3D Vision 3DV 2020
Xuan Luo, Yanmeng Kong, Jason Lawrence, Ricardo Martin-Brualla, and Steven M. Seitz

Slow Glass: Visualizing History in 3D | Fourth Workshop on Computer Vision for AR/VR CVPR-W 2020
Xuan Luo, Yanmeng Kong, Jason Lawrence, Ricardo Martin-Brualla, and Steven M. Seitz

Pepper's Cone: An Inexpensive Do-It-Yourself 3D Display UIST 2017
Xuan Luo, Jason Lawrence, and Steven M. Seitz

Purine: A Graph-based Deep Learning Framework ICLR 2015
Min Lin, Shuo Li, **Xuan Luo**, and Shuicheng Yan

Honors and Awards

Best Paper Award, SIGGRAPH Asia 2024
Selected EECS Rising Star by EECS at UC Berkeley Nov. 2020
Anne Dinning - Michael Wolf Endowed Regental Fellowship, UW 2015 – 2016
Distinguished Undergraduate Scholarship, SJTU 2015
Shanghai Outstanding Graduate, Shanghai 2015
National Scholarship, China | *Highest scholarship in China* 2013
2012 University Physics Competition, Silver Medal, USA 2012

Media & Press

Time-Travel Rephotography 2020
Two Minute Papers, GIZMODO, Hack a Day, QubitAI, Guokr, Tencent, marktechpost.com, TechTheLead,
Tech Times, The Science Times, Review Geek, Tech Explore, msnice.net, kknews.cc

Consistent Video Depth 2020
Two Minute Papers, QubitAI, Synced, Medium, Bo Yu AI

Pepper's Cone 2018
"Demo Hour" of *ACM Interactions Magazine*, iProgrammer, Hack a Day, Hacker News

Professional Service

WiGRAPH: Women in Graphics Research (wigraph.org) Fall 2020 – 2025
Rising Stars Coordinator

Poster Juror
- The ACM Special Interest Group on Computer Graphics (**SIGGRAPH**) 2025